



1. Description

1.1. Project

Project Name	Gcraud_shuffle_IR
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	06/09/2026

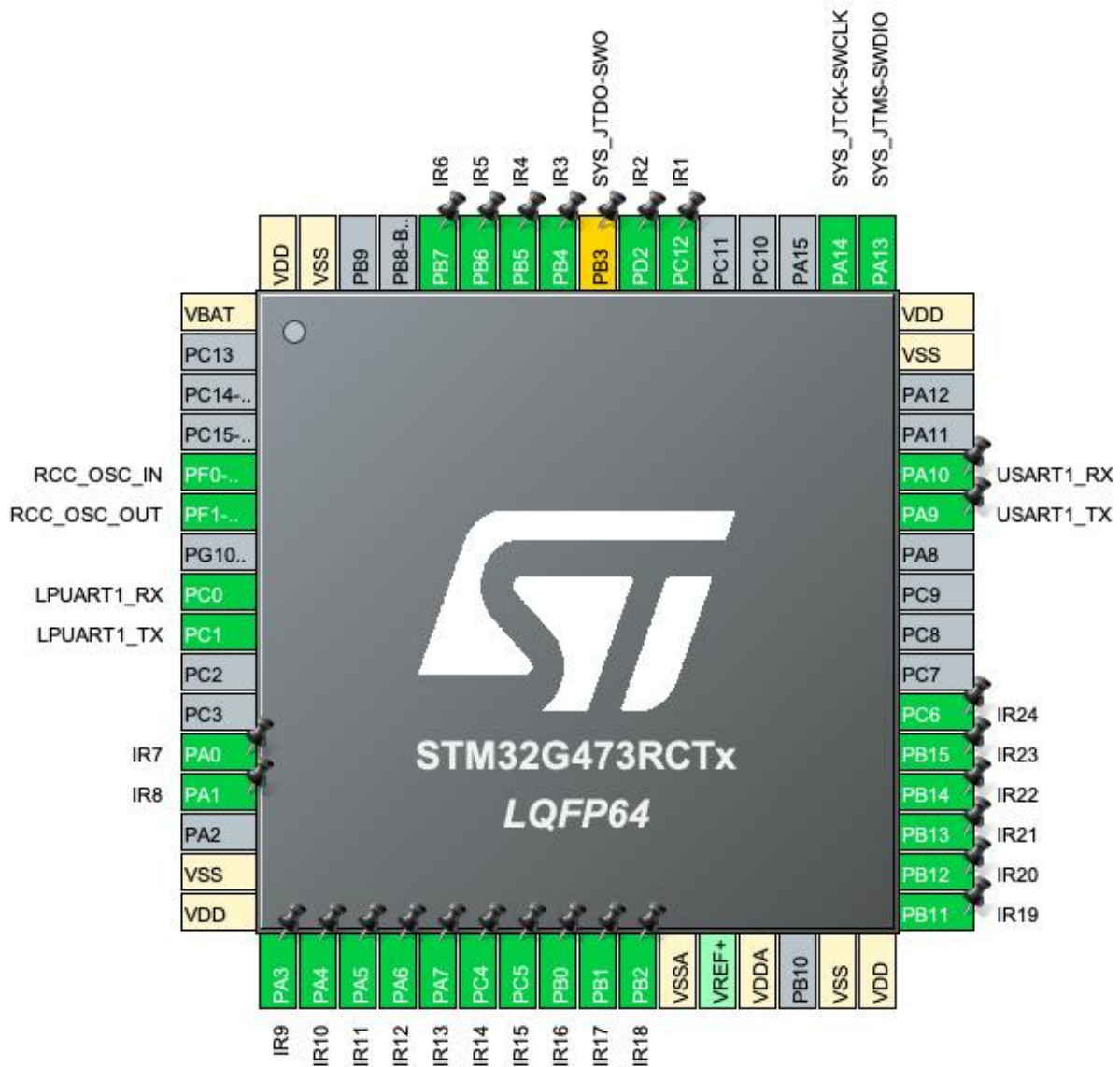
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x3
MCU name	STM32G473RCTx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

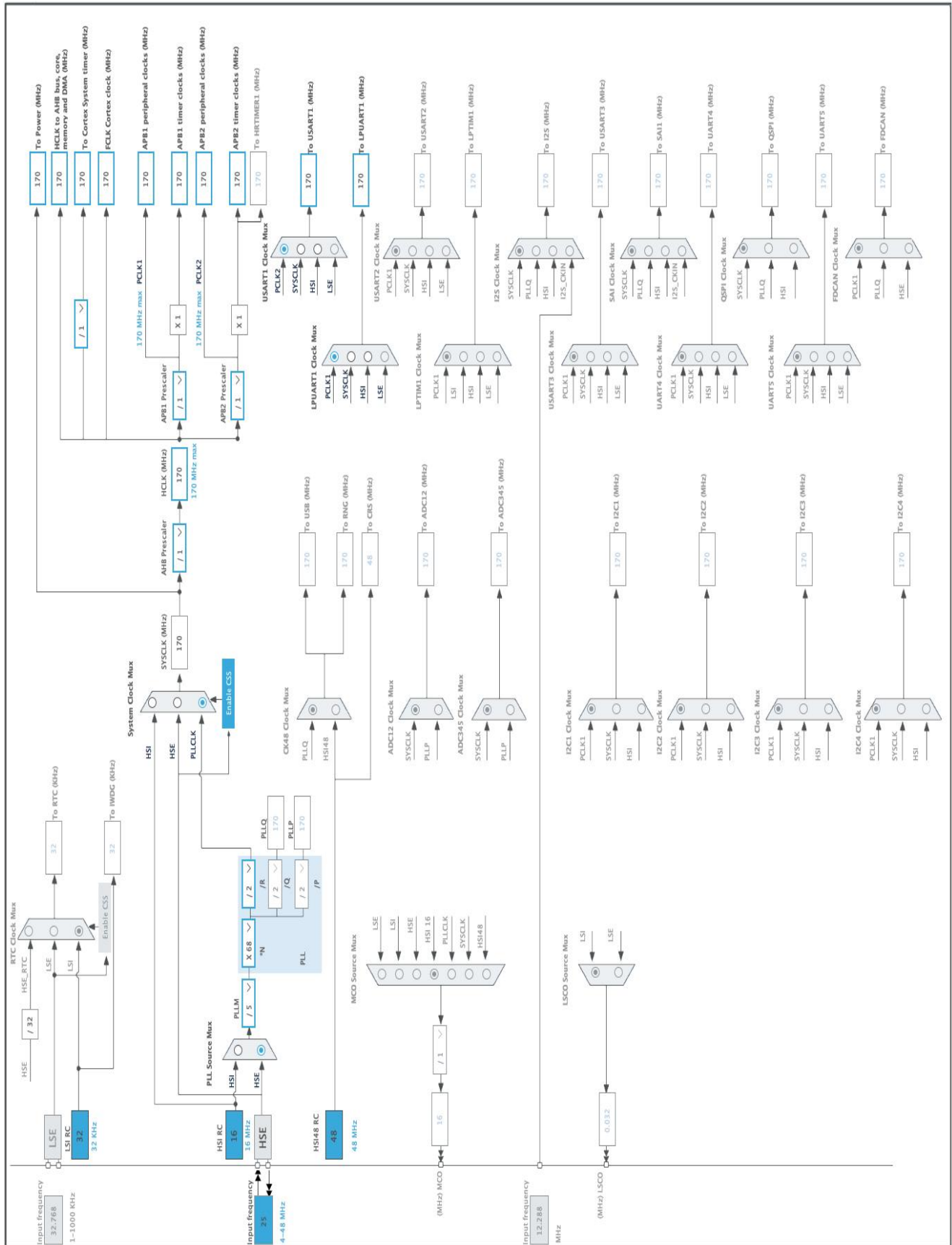
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
5	PF0-OSC_IN	I/O	RCC_OSC_IN	
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	
8	PC0	I/O	LPUART1_RX	
9	PC1	I/O	LPUART1_TX	
12	PA0 *	I/O	GPIO_Input	IR7
13	PA1 *	I/O	GPIO_Input	IR8
15	VSS	Power		
16	VDD	Power		
17	PA3 *	I/O	GPIO_Input	IR9
18	PA4 *	I/O	GPIO_Input	IR10
19	PA5 *	I/O	GPIO_Input	IR11
20	PA6 *	I/O	GPIO_Input	IR12
21	PA7 *	I/O	GPIO_Input	IR13
22	PC4 *	I/O	GPIO_Input	IR14
23	PC5 *	I/O	GPIO_Input	IR15
24	PB0 *	I/O	GPIO_Input	IR16
25	PB1 *	I/O	GPIO_Input	IR17
26	PB2 *	I/O	GPIO_Input	IR18
27	VSSA	Power		
29	VDDA	Power		
31	VSS	Power		
32	VDD	Power		
33	PB11 *	I/O	GPIO_Input	IR19
34	PB12 *	I/O	GPIO_Input	IR20
35	PB13 *	I/O	GPIO_Input	IR21
36	PB14 *	I/O	GPIO_Input	IR22
37	PB15 *	I/O	GPIO_Input	IR23
38	PC6 *	I/O	GPIO_Input	IR24
43	PA9	I/O	USART1_TX	
44	PA10	I/O	USART1_RX	
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	
50	PA14	I/O	SYS_JTCK-SWCLK	
54	PC12 *	I/O	GPIO_Input	IR1

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
55	PD2 *	I/O	GPIO_Input	IR2
56	PB3 **	I/O	SYS_JTDO-SWO	
57	PB4 *	I/O	GPIO_Input	IR3
58	PB5 *	I/O	GPIO_Input	IR4
59	PB6 *	I/O	GPIO_Input	IR5
60	PB7 *	I/O	GPIO_Input	IR6
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x3
MCU	STM32G473RCTx
Datasheet	DS12712 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

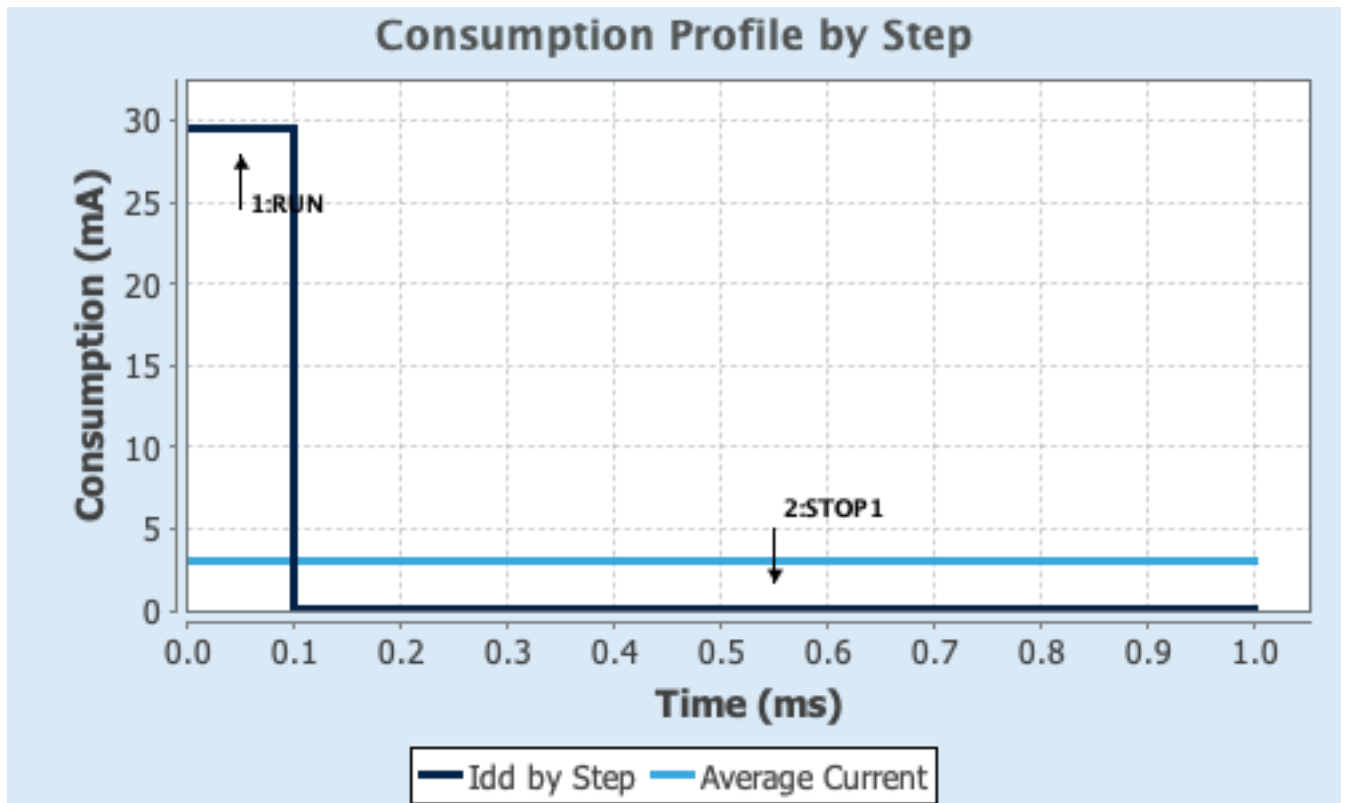
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	Gcraud_shuffle_IR
Project Folder	/Users/takum/RCJJ/Gcraud-Shuffle/japanopen/git/2026_IR/Gcraud_shuffle_IR
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_G4 V1.5.2
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_LPUART1_UART_Init	LPUART1
4	MX_USART1_UART_Init	USART1

3. Peripherals and Middlewares Configuration

3.1. LPUART1

Mode: Asynchronous

3.1.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.2. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.2.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
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HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost
Peripherals Clock Configuration:	
Generate the peripherals clock configuration	TRUE

3.3. SYS

Debug: Serial Wire

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.4. USART1

Mode: Asynchronous

3.4.1. Parameter Settings:

Basic Parameters:

Baud Rate	460800 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
LPUART1	PC0	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC1	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
Single Mapped Signals	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	
GPIO	PA0	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR7
	PA1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR8
	PA3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR9
	PA4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR10
	PA5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR11
	PA6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR12
	PA7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR13
	PC4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR14
	PC5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR15
	PB0	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR16
	PB1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR17
	PB2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR18
	PB11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR19
	PB12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR20
	PB13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR21
	PB14	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR22
	PB15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR23
	PC6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR24
	PC12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR1
	PD2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR2
	PB4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR3
	PB5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR4
	PB6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR5

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IR6

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	true	0	0
LPUART1 global interrupt	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	false	true	true
LPUART1 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Utilities

DMA

GPIO 

NVIC 

RCC 

SYS 

LPUART1 

USART1 

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstpfc11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-

digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-use-tips-and-recommendations-stmicroelectronics.pdf
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- Application Notes https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-

usb-typec-power-delivery-for-stm32-mcus-and-mpus-
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eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4894-how-to-use-
eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf)

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oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-
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oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-
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thermal-management-on-stm32-applications-stmicroelectronics.pdf)

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fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5405-how-to-use-
fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf)

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mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-
stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5978-introduction-to-
mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-
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vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5690-how-to-use-
vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)

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random-number-generation-validation-using-the-nist-statistical-test-suite-
for-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4230-introduction-to-
random-number-generation-validation-using-the-nist-statistical-test-suite-
for-stm32-mcus-and-mpus-stmicroelectronics.pdf)

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dma-controller-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an2548-introduction-to-
dma-controller-for-stm32-mcus-stmicroelectronics.pdf)

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timers-for-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4013-introduction-to-
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pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-
mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4277-how-to-use-
pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-
mcus-stmicroelectronics.pdf)

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optimize-lpuart-power-consumption-on-stm32-mcus-
stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4635-how-to-
optimize-lpuart-power-consumption-on-stm32-mcus-
stmicroelectronics.pdf)

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using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-
tamp-with-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4759-introduction-to-
using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-
tamp-with-stm32-mcus-stmicroelectronics.pdf)

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with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf

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